

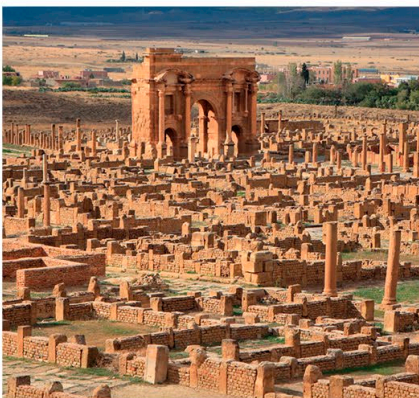
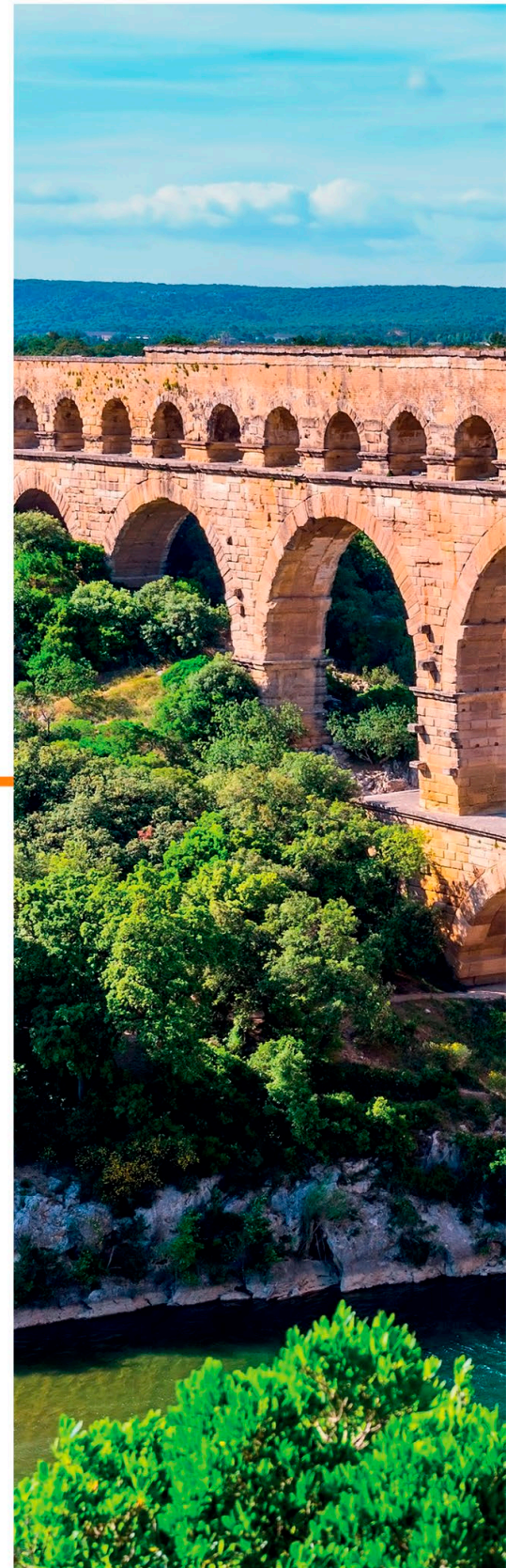
509 BCE–395 CE

ROMAN ENGINEERING AND ARCHITECTURE

The Romans were inventive engineers and relentless builders. Of the buildings that survive from the Republican era (before 27 BCE), many are temples, with high podia and Greek-style columns, but these are complemented by public buildings, such as the rectangular column-aisled basilicas used for law courts and audience chambers. From the Imperial era, Roman architecture developed its own clear identity, relying on new materials such as brick and “Roman concrete” (*opus caementicium*), which incorporated volcanic pozzolana sand for hardness and

durability. This allowed the building of domes, huge amphitheatres such as the Colosseum, aqueducts, and great baths, including the Baths of Diocletian in Rome.

The Romans decorated their buildings with carved friezes, portrait busts, and fine statues in marble and bronze, often based on Greek or Hellenistic originals. Roman interiors were painted in stucco and fresco with colourful scenes, both geometric and mythological. Villa floors and walls were adorned with mosaics, often of arena or legendary scenes, made up of thousands of marble and limestone tesserae.



KEY MOMENTS

c. 300 BCE–395 CE Roman roads

The Romans built a network of 80,000 km (50,000 miles) of paved roads (such as the *Appian Way*, left) linking major points in the empire. Roads were built straight on levelled earth; layers of gravel were added and topped with a hard concrete surface and blocks of rectangular stone. A pronounced camber and adjacent ditches kept them well drained.

c. 200 BCE–395 CE Roman grid cities

Cities founded by the Romans were laid out on a grid pattern based on the plan of a Roman military camp. A main street, the *decumanus maximus*, ran east-west and another, the *cardo maximus*, north-south, with major buildings and the forum, the centre of civic life, placed where they intersected. To lay out the cities, Roman surveyors, or *agrimensores*, used ropes, rods, and the *groma*, a pole with a rotating arm and suspended weights.